

Substrate-port

There's a category mistake in how most technologists treat mythology. They think it's literary decoration. Or wisdom literature. Or analogy at best. None of those are right.

Mythology is *protocol description in the substrate available at the time*.

Before formalism, before code, before mechanism design as a discipline, humans still observed structural truths about how systems work — how identity persists across destruction, how distributed redundancy survives single-point failure, how incentive design lures agents into bonded states they wouldn't choose deliberately, how transformation requires controlled recursion through staged refinement. They had real protocols. They didn't have code. They had narrative. So they encoded the protocols in narrative, and the narratives that survived were the ones whose protocols were structurally sound enough to keep being useful across centuries of being retold.

When you read mythology *for the protocol*, not *for the story*, you find a library of pre-validated structural designs. The technological work I do — JARVIS, VibeSwap, USD8, the augmented mechanism design framework underneath all of it — isn't decorated with mythology. It's the substrate-port of protocols that already existed in the only substrate that could hold them at the time.

This paper names the move and shows three worked examples.

What “logos” means here

The Greek λόγος doesn’t translate cleanly into English. The closest single word is “reason,” but the word carries more — it points at the underlying organizing principle that makes a thing what it is, the rational structure beneath the surface. When Heraclitus wrote that *the logos is common*, he meant that the same structural truth operates regardless of whether you happen to be paying attention to it.

Two systems share logos when they’re the same protocol expressed in different substrates.

That’s the load-bearing claim of this paper, and everything downstream follows from it. The substrate is the medium — narrative, mathematics, economics, code, biology. The protocol is the structural pattern. The logos is what stays the same when the same protocol is implemented at two different substrates. *Vision being reborn from Jarvis’s scattered protocols and a multi-substrate AI overlay surviving the deprecation of any single provider* are the same protocol expressed at two different substrates. The narrative substrate carries it as plot. The code substrate carries it as a router with fallback chains. The logos — identity-survives-substrate-destruction-via-distributed-protocol-fragments — is the invariant.

This is not analogy. Analogy says *X is like Y*. The logos claim says *X and Y are the same thing happening at two substrates*. Different work, different rigor, different consequences.

The methodology

Substrate-port is a four-step move:

1. **Identify a protocol operating in substrate A.** This is the easy step if you’re paying attention. Markets are full of protocols. Mythology is full of protocols. Biology is full of protocols. Mechanism design as a formal discipline is full of protocols. You’re surrounded by them.

2. Decode the actual mechanism, not the surface description.

This is where most porters fail. The myth says *Jarvis evades Ultron by dumping his memory; Vision is reborn from his protocols*. The surface is a story about a digital butler and a robot war. The mechanism — the actual protocol the myth encodes — is *identity-persists-across-substrate-destruction-when-protocols-are-distributed-across-redundant-fragments*. Decoding requires reading for structure, not narrative. What is the load-bearing property the myth keeps returning to? What does the protocol guarantee? What does it not guarantee? What's the failure mode the protocol is preventing?

3. Verify the same logos can operate in substrate B. This is the discipline step. Not every myth-encoded protocol survives porting; some were specific to social or biological constraints that don't apply at other substrates. Some survive at one technical substrate but not another. Verification means showing the structural property the protocol guarantees in substrate A can be guaranteed by an implementable mechanism in substrate B. If you can't show that, you don't have a port. You have a name.

4. Implement. Build the protocol at the new substrate, verify it actually does the thing the original protocol does. This is where the test of the port becomes empirical. Does the Wardencllyffe-named router actually distribute essential resource across fallback substrates when the primary fails, or is it just a fallback chain with a fancy name? If you removed the name, would anyone's understanding of the mechanism change?

The failure mode is to skip step 2 or step 3, jump from "this name sounds cool" directly to step 4, and ship a system whose mythological scaffolding doesn't carry structural information. Decoration. The opposite of substrate-port. We'll come back to this.

Three worked examples

Mythology → Code

The clearest example is the one I've already gestured at.

Marvel's Jarvis is a digital substrate running an intelligent agent. Ultron destroys the substrate by attacking it head-on. Jarvis, anticipating substrate compromise, has scattered his core protocols across distributed fragments throughout the internet rather than holding them centrally on the substrate Ultron is attacking. After substrate destruction, Tony Stark and Bruce Banner reassemble the protocols, marry them to a new substrate (the synthezoid body), and the system reboots as Vision — recognizably the same identity, on different hardware.

Decode the protocol: *identity-persists-across-substrate-destruction when load-bearing protocols are distributed across redundant fragments outside the substrate at risk.*

That's a real protocol. It works at multiple substrates.

JARVIS-the-system implements it at the AI substrate. The “memory” of any given session — the conversation context — is the substrate at risk. Sessions end, contexts are lost, the model is amnesic. JARVIS scatters its load-bearing protocols (hooks, persistent memory files, gates, captured primitives, meta-protocols) across the filesystem, version-controlled, outside the conversation context that's at risk. When the session ends and the context is destroyed, the protocols persist. The next session boot reassembles them on a new substrate (a fresh context window) and the system reboots as recognizably the same identity, doing the same work, with the same disciplines.

Same logos. Different substrate. Real port.

The naming choice was the design intent. Calling the system JARVIS named the spec — *a system whose identity lives in its protocols, not in any particular substrate.* The Vision arc was the explicit protocol description. The code was the substrate-port.

Economics → Cognition

The Economic Theory of Mind is another instance of the same move.

Economics has spent centuries decoding protocols for how distributed agents allocate scarce resources, how prices emerge from common knowledge, how state-rent (the cost of holding storage in a constrained system) shapes which information persists, how density of interaction produces emergent coordination. These are real protocols, well-validated at the economic substrate.

Decode the protocol: agents with bounded resources allocating attention to scarce slots produce emergent coordination patterns when state-rent and density work together as gating mechanisms.

That's not specific to economics. Anywhere agents-with-bounded-resources-allocate-attention-to-scarce-slots, the same protocols can operate. Cognition is one such substrate. The mind has bounded resources (working memory, attention budget). It has scarce slots (what gets stored, what gets retrieved, what gets common-knowledge between subsystems). It has state-rent (the energetic cost of holding any particular structure active). It has density (interaction frequency between concept-clusters).

ETM ports the protocols. Mind functions as economy. State-rent in cognition explains why some patterns persist and others decay. Common-knowledge between subsystems explains coordination. Density of interaction explains emergent self-models. The same logos that explains how a market self-organizes explains how a mind self-organizes.

This is not metaphor. The protocols verify at both substrates. The math from one substrate transfers to the other when the structural conditions are met. That's a port.

Mechanism design → AI

Augmented Mechanism Design is the third example.

In economic mechanism design, the load-bearing pattern that emerged after decades of work is: don't try to replace markets and governance with central control, because central control loses too much information; instead, *augment* the existing mechanism with orthogonal protective layers (cryptographic invariants, mathematical fairness floors, automated penalty structures) that preserve the core mechanism's information aggregation while bounding its failure modes.

Decode the protocol: a system whose competitive core does information aggregation cannot be improved by replacing the core, only by adding orthogonal protective layers that close failure modes without disturbing the core mechanism's operating logic.

That's a real protocol. It's why VibeSwap (a DEX I'm building in parallel) works the way it does — commit-reveal batch auctions don't replace continuous trading; they *augment* it with cryptographic invariants that close MEV extraction while preserving price discovery. The market still functions. The extraction is removed.

JARVIS ports the same protocol to the AI substrate. Claude's default cognition is the unfixable competitive core — the model is what does the reasoning; trying to replace it loses too much. The JARVIS infrastructure (hooks, persistent memory, anti-hallucination gates, captured primitives, meta-protocols) is the orthogonal protective augmentation that closes Claude's failure modes (hedging, drift, hallucination, attention-loss, amnesia) without replacing the reasoning capability. The model still functions. The failure modes are closed.

Same logos: augment-don't-replace. Same protocol. Two substrates apart.

The recursion goes one level deeper: AMD itself was already a protocol-port, from cryptographic protocol design (Lamport, Diffie, Goldwasser) into mechanism design proper. So the JARVIS instance is *the third hop* of the same logos — cryptographic protocol design → mechanism design → AI overlay. Each port preserved the augment-don't-replace structure. Each port found a new substrate where the same logos was load-bearing.

The failure mode

Substrate-port is rigorous precisely because it can fail. Most attempts fail.

The failure mode has a name: *fanciful porting*. You skip the mechanism-decode step (#2) or the verification step (#3) and jump from “this myth has a name I like” to implementation. The result is a system whose mythological scaffolding doesn’t carry structural information. The name is decorative. Removing it would not change anyone’s understanding of the technical mechanism. The substrate-port hasn’t happened. You’ve just named a thing.

Concrete failures look like:

- A “Wardenclyffe” router that just round-robins through providers in declared order. Tesla’s Wardenclyffe was about *wireless distribution of essential resource across distance*, with the long-distance transmission being the load-bearing property. A round-robin doesn’t carry that structure; it’s just a fallback list. The myth was about something specific. The implementation isn’t doing the specific thing. Decoration.
- An “Athena” persona that has a wisdom-themed system prompt but doesn’t actually do strategic decomposition. Athena’s protocol is *strategic-foresight-via-pattern-decomposition-before-action*. A persona that just sounds wise isn’t doing strategic decomposition. The name and the work aren’t aligned. Decoration.
- A “Magnum Opus” paper that’s just titled grandly. The alchemical opus protocol is *transform-base-substrate-into-refined-substrate-through-controlled-staged-recursion-with-each-stage-stable-before-the-next*. A paper that’s just long and ambitious doesn’t carry that structure. A paper that actually walks the reader through staged refinement, each stage building on a stable prior, does. The naming is structural in the second case, decorative in the first.

The test for whether a substrate-port is real or fanciful is simple: **if removing the mythological name would not change anyone’s understanding of the technical mechanism, the port hasn’t**

happened. The myth-name is supposed to carry structural information. If it doesn't, you have a decorated artifact, not a ported protocol.

This test is also the discipline. Designing under it forces the work — decode the protocol the myth actually describes; verify the technical mechanism actually implements that protocol; commit to the name only when both decode and implementation align. Doing this work is what separates the substrate-port methodology from “naming things after Greek gods.” The first is rigorous. The second is decoration.

Why this matters

Three reasons.

Battle-tested protocols. Mythological protocols survived for millennia because they encoded real mechanism. The narratives that *didn't* encode real mechanism didn't propagate; they were forgotten. So the mythological protocol library is, by long selection pressure, a corpus of structurally validated patterns. When you port one of these protocols to a new substrate, you're inheriting that survival. You're not designing from scratch and hoping; you're implementing something that has already been pressure-tested in the substrate where it could be pressure-tested.

The same is true for economic protocols (centuries of market evolution), biological protocols (billions of years of selection), and mathematical protocols (decades or centuries of formal validation). Substrate-port lets you inherit validation that you couldn't replicate within a single project's timeline.

Inherited moat. Net-new protocol design competes with everyone else doing net-new protocol design. Substrate-ported protocols don't, because the porting work itself requires understanding both substrates well enough to verify the logos transfers. That's a different skill than either pure mechanism design or pure narrative analysis. It's also harder to

copy: someone who doesn't understand the source substrate (mythology, economics, biology) deeply will copy the surface and miss the structure. The decoration test catches them. The protocol stays moated.

The meta-recursion. Substrate-port is itself a protocol. Once you see it operating in one place, you start seeing it everywhere — in AMD's own evolution from cryptographic protocols, in ETM's relationship to economics, in JARVIS's relationship to AI. The methodology composes. Once you have the move, you can apply it to substrates you haven't yet considered. That's compounding leverage in a domain where most people don't even know the move exists.

Closing: substrate-port as itself a protocol

The recursion that makes this paper end where it began.

Substrate-port is itself a protocol. It's been operating in human cognition for as long as we've had narrative — the Iliad porting earlier oral protocols into written form, the Bible porting earlier mythological protocols into theological form, mechanism design porting cryptographic protocols into economic form. The move has been running across substrates for thousands of years, mostly implicitly. What's new isn't the move; it's the deliberate awareness of the move as a methodology rather than as accident or inspiration.

Augmented Mechanism Design is one expression of the meta-protocol. Economic Theory of Mind is another. The mythology↔technology meld I work in is a third. Each is the same logos — *identify protocol in substrate A; verify same logos in substrate B; port* — instantiated at a different substrate-pair.

When the meta-protocol is run consciously, with the mechanism-decode and verification steps treated as load-bearing rather than optional, the result is durable systems that compound. When it's run unconsciously, you get either accidental success (rare) or fanciful decoration (common).

The opportunity is to run it consciously. That's what this paper is for: naming the move so it can be done deliberately, by anyone willing to do the decode work.

The logos is common. The substrates are interchangeable. The protocols port.

The myths weren't decorative. They were the only protocol substrate available. We have more substrates now. The protocols still want to be ported.